

Ecolight Maestro

Time & Light Intensity Based Street Light Automation using LPC2129 ARM7

Project Report

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1. Project Overview

Ecolight Maestro is an intelligent street light automation system developed using the LPC2129 ARM7 microcontroller. The system automatically controls street lights based on real-time clock scheduling and ambient light intensity detected via an LDR sensor.

Traditional street lighting systems often remain ON unnecessarily, leading to excessive energy consumption. Ecolight Maestro addresses this issue by combining time-based automation with light intensity monitoring, ensuring that street lights operate only when required, resulting in an energy-efficient and fully automated smart lighting solution.

2. Objectives

- Reduce power wastage in street lighting systems
- Automate street light operation based on environmental conditions
- Provide user-friendly RTC configuration
- Demonstrate ARM7 peripheral integration
- Implement interrupt-driven embedded system design

3. Key Features

3.1 Real-Time Clock Management

- Displays current time, date, and day
- Supports RTC parameter modification via keypad

3.2 Intelligent Light Detection

- LDR-based ambient light sensing
- ADC conversion for light intensity measurement
- Automatic day/night detection

3.3 Automatic Street Light Control

- Lights turn ON during low-light conditions
- Lights turn OFF during daylight conditions
- Dual-condition verification using RTC and LDR

3.4 User Configuration Interface

- 4x4 matrix keypad for RTC parameter editing
- Menu-driven operation

3.5 Interrupt-Driven Design

- EINT0 external interrupt support for instant menu access
- Responsive user interaction

3.6 LCD Monitoring

- Displays time, date, day, ADC value, menu options, and system status

4. Hardware Components

Component	Purpose
LPC2129 ARM7 MCU	Main Controller
LCD 16x2	User Interface
LDR Sensor	Light Detection
ADC Channel	Sensor Data Acquisition
RTC Module	Time Management
4x4 Keypad	User Input
LEDs	Street Light Simulation
Push Button	EINT0 Interrupt Trigger
Resistors	Current Limiting & Pull-ups

5. Software Tools

- Keil uVision
- Proteus (Circuit Simulation)
- Flash Magic
- Embedded C
- Git & GitHub

6. Working Principle

The system continuously performs the following operations in a loop:

- Step 1 - Initialize: LCD, RTC, ADC, GPIO, and Interrupts
- Step 2 - Read current Hour, Minute, Second, and Date from the RTC
- Step 3 - Read the LDR value through the ADC
- Step 4 - Determine whether the environment is Daytime or Nighttime, based on RTC time and light intensity
- Step 5 - Control the street lights automatically based on that decision
- Step 6 - Monitor the external interrupt (EINT0); when triggered, open the RTC configuration menu, allow parameter modification via the keypad, and save the updated values

7. Street Light Decision Logic

LED OFF Condition

Time < 18:00 OR ADC Value > 300

Example: Time = 14:00:00 -> Result: LED OFF

LED ON Condition

Time >= 18:00 AND ADC Value <= 300

Example: Time = 19:30:00, ADC Value = 180 -> Result: LED ON

8. System Flow

START -> Initialize LCD -> Initialize RTC -> Initialize ADC -> Read Time -> Read LDR -> Evaluate (Time >= 6 PM AND ADC < Threshold?) -> If YES: Turn LEDs ON, else Turn LEDs OFF -> Update LCD Display -> Check EINT0 Interrupt -> If triggered: Open RTC Config Menu, else Continue Loop -> Repeat

9. RTC Configuration Menu

The RTC settings can be modified using the keypad whenever EINT0 (P0.16) is triggered. Available menu options:

- 1 - Update Hour
- 2 - Update Minute
- 3 - Update Second
- 4 - Update Day
- 5 - Update Date
- 6 - Update Month
- 7 - Update Year
- 8 - Exit Menu

10. Applications

- Smart Street Lighting
- Smart Cities
- Campus Lighting
- Industrial Lighting

11. Future Enhancements

- Wireless Monitoring using IoT
- GSM-Based Fault Reporting
- Solar Powered Street Lights
- Motion Detection for Adaptive Brightness
- Cloud-Based Energy Analytics
- Mobile Application Integration

12. Learning Outcomes

Through this project, practical experience was gained in:

- ARM7 LPC2129 Programming
- Embedded C Development
- RTC Programming
- ADC Interfacing
- Sensor Integration
- Interrupt Handling

13. Conclusion

Ecolight Maestro demonstrates a practical, low-cost approach to automated street lighting by combining RTC-based time scheduling with real-time ambient light sensing on the LPC2129 ARM7 platform. The dual-condition decision logic ensures lights operate only when genuinely needed, the keypad-driven RTC menu gives users direct control without reprogramming the device, and the interrupt-driven design keeps the system responsive. The project validates core ARM7 peripheral integration skills - GPIO, ADC, RTC, and external interrupts - and establishes a solid foundation for future IoT and solar-powered extensions.

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